GENSET

625 kVA LIQUID (RADIATOR) COOLED

INSTALLATION GUIDELINE
&
OWNER'S MANUAL

Published on behalf of Customer Support, Kala Genset Pvt Ltd , Pune -411 035. INDIA

KALA GENSET PVT LTD.

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Overall Specifications of KOEL Green Liquid (Radiator) cooled Silent Gensets

Genset Model	Engine Model	Engine Power at 1500 rpm. kW (hp)	Prime Power Genset rating at 0.8 pf (kVA)	Overall Canopy Dimensions L x W x H (mm) Tol $\pm$ 50	Approx. weight (with canopy) kg.
KG1-625WS	DV12TA G2	552 (750)	625	6660 x 2000 x 3484*	7800

* Including silencer

• **Genset ratings** are as per ISO 8528 performance class G3. • **Prime Power Rating** is the maximum power available continuously for a variable electrical load for unlimited number of hours per year under standard operating conditions. • **Specific gravity of diesel** is considered as 0.845 (+5 % tolerance as per ISO 3046) .

Sizing of KGPI Genset

To select the right generating set under a given set of circumstances, knowledge of generating set and its performance under various conditions is necessary. It is also equally necessary to understand load characteristics and to make a load analysis. Beyond that, the effect of the voltage regulation systems plays an important role.

It is better to select a slightly larger generator capacity than the available load; the reason is that additional uses are invariably observed after it has been installed. These are usually not predictable. By having a margin available it is possible to power some loads which may have been overlooked during Genset selection, or which may be added after the installation of the Genset

Ratings

The rating of an alternator is expressed either in kilowatt (kW) or kVA. The relation between kW and kVA rating is :
kW = kVA x Power Factor.

kW = 1.732 x V x A x Power Factor (for 3 phase alternator) / 1000

kW = V x A x Power Factor (for a single phase alternator) / 1000

Where V = Line voltage; A = Line current

Load Variation between phase should not more than 10% and min 50% load should be available at the time of commissioning

Power Factor

The power factor of electrical system depends upon the nature of characteristics of load e.g Induction motors, Furnaces, thyristor, power electronics etc. Power Factor is lagging for inductive load. Power Factor is leading for capacitive load. Power factor is unity for purely resistive loads. Normally, a generator is designed for a power factor of 0.8 The table below gives power factors of non-motor loads

Type of load	Power Factor
Incandescent lights	1.00
Heating elements, ovens	1.00
Fluorescent lights	0.90
Induction furnaces	0.60 - 0.70
Arc furnaces	0.80 - 0.90
Welding Set Transformer)	0.60
Welding Set(AC Motor-DC Generator) & Transformers	0.80 - 0.90

- Load variation between phases should not be more than 10%.
- KVA x 1.39 = AMP per Phase for III phase gen set & KVA X 1.39 x 3 AMP for I phase gen set at 100% load . Minimum 50% of AMPs per phase is required at the time of commissioning.

Deration Factor

The engine's horse power as mentioned on its name plate is the HP which an engine can deliver in MSL at NTP conditions. HP delivered at site will vary depending on site conditions (deration due to altitude , temperature and humidity) for details of site derations please consult seller or KOEL area office.

Input HP (Metric) = kW / 0.736 x Alternator Efficiency.

Load Calculation

Steps to be followed while sizing the Genset.

Step 1:
Collect all necessary basic data of load. Tabulate the details such as lighting load, motor load, A/C load, Heaters load etc with quantity and name plate details.
For motor loads, mention starting method also, i.e. Type of starter used.

Step 2 :
Applicable For Three Phase Genset Convert all single phase loads to three phase loads by distributing equally on three phases.

Step 3:
Starting power (kVA) of each equipment is based on type of starting method.
For eg. In case of motors,
Starting by DOL will pull up to 6 to 7 times the rated full load kW of a motor.
Star/Delta will pull up to 2 to 3 times the full load kW of the motor.
Soft Start will pull up to 1.2 to 1.5 times the full load kW of the motor. Rectifier loads may also pull up to 50% above their rated KVA.

Step 4:
Sum up all the steady state running loads in kVA and sum up all the starting kVA's . These two figures will be used to size up the Genset and sequence of loading.

Step 5:
Determine the maximum allowable voltage dip which the Genset should experience when the total starting KVA is applied on this Genset. This voltage dip percentage is usually known to the customer based on equipment type to be powered.

Step 6:
The previously obtained figures (Running KVA, Starting KVA, Voltage, Voltage Dip, Loading sequence) will decide Genset size.

Installing KGPL Genset

Location

Dust and Fumes are the greatest danger to generating set as they could lead to clogging of cooling and air system which may affect Genset performance.

- **If KGPL Genset is to be installed in Free field condition** , please ensure dust free location and clear space of 2 times the height of Genset.
- **If KGPL Genset is to be installed in the room**, please ensure proper cross ventilation, clear space of minimum 3 meters from all sides.
- **If KGPL Genset is to be installed on a roof top**, please ensure structural clearance, proper cross ventilation with respect to wind direction, clear space of minimum 3 meters from all sides
- **If KGPL Genset is to be installed in the basement**, please ensure ventilation with respect to Air requirement and clear space for easy maintenance.

Layout

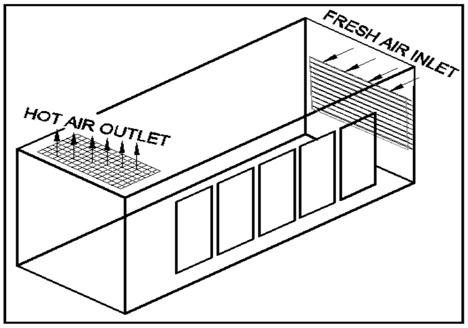
- **Allow easy accessibility** to Genset for easy operation and maintenance.
- **Locate the Genset** considering the prevailing wind direction.
- **Genset installation** should be as close as possible to the load. However in some cases , like stone crusher plant, Garment plant, Cement factory, Paint factory, chemical factory, etc the Genset should be isolated from the dust, hazardous and chemical fumes contained in the surrounding air.
- **In case, Genset is to be installed in heavy dusty conditions**, it is recommended to use heavy duty air cleaners (Please contact Seller or KOEL Area office for further guidelines).
- **For fuel supply and fuel storage arrangement** for Genset , follow the local rules and regulations.
  * The environment protections . * The fire protection .
- **In case of high humid atmospheric conditions**, an anti condensation heater should be used for the Alternator.
- **A rooftop installation** requires proper planning and the structural design considerations for dynamic loading.
- **Proper consideration should be given to take exhaust gases away** from Genset location and shielded for hot surface.

Noise

- There should be **sufficient space** around the Genset to avoid resonance and echo effect which contributes to abnormal sound from Genset.
- The space should be open without any obstacles.

Ventilation

- Cross ventilation and free flow of cool, clean and fresh air
- The space should be open without any obstacles.
- For room and basement installations, supply of fresh air and removal of hot air should be done through forced ventilation with suitable air ducts.
- Maximum allowed temperature rise above ambient in Genset Enclosure at air intake of engine is 7–9 °C



Foundation

Foundation is one of the important factors affecting the successful operation and life of a generating set. Improper foundation may result in alignment and vibration problems which may subsequently lead to pre-mature failure of Genset Components and Safety.

The basic functions of foundation are given below.

- To provide leveled platform to seat and seal the Genset. • To support the weight of the entire generating set.
- Support the dynamic load of Genset while running. • Absorb the vibration produced by the rotating and reciprocating masses.• Isolate the surrounding structures by absorbing the vibration of generating set while running.

Refer Attachment for Foundation Plan Details & recommendations.

While casting the concrete foundation, the bearing strength of the soil at site of installation should be determined. The table below gives approximate safe bearing capacity of various soils for ref.

- **The length and width of foundation should be at least 300 mm more** on each side than acoustic enclosure length and width respectively.
- It is recommended to have **foundation height of 150 to 200 mm** above ground level to maintain cleanliness and avoid flooding.
- **Check the foundation level diagonally** as well as across the length and width for **even flatness** and same should be within $\pm$ 50 mm in horizontal plane in any one direction only.
- **Ensure that the concrete is completely cured** before positioning the enclosure.
- **Ensure that the foundation to support 1.5 times of the total wet weight** of the single generating set installation and 2 times of the total wet weight for the multiple generating set installation.
- **For rooftop installations,** it is necessary to have planning and structural design approval.

Do not install acoustic enclosure on loose sand or clay. **Avoid hard / sharp projections** such as stone or steel parts on foundation surface, which may damage fuel tank at the bottom.
Avoid uneven foundation which may lead to improper resting of Genset on foundation, there by leading to leakage of sound and vibration on Genset.
If Genset is to be mounted on foundation having uneven surface under unavoidable circumstances, use appropriate thickness of rubber matting above foundation for positive sealing of Genset with base.

Unloading and Positioning

- All the Gensets are provided with suitable lifting arrangements. unload the Genset, by lifting with proper lifting tackle/nylon sling, /soft sling with spreader bar/spacer plate of suitable capacity to avoid any damage
- **Do not drag the Genset on floor/ground.**
- **Use lift points on the acoustic enclosure** (canopy) which provides support to the equipment when it is moved

Exhaust System

Genset Exhaust system directs the hot flue gases into the atmosphere harmlessly.

On KGPL Genset, exhaust silencer and expansion bellow are the integral parts of the exhaust system. At the time of Genset installation, **please ensure proper alignment of exhaust silencer with respect to expansion bellow**.

While taking exhaust gases away from Genset above stack height, please refer to the table below for selecting the size of exhaust pipe.

As per Statutory regulations, exhaust pipe has to be extended above the surrounding stack as below.

H = h + 0.2 x √ kVA

Where H = Minimum height of exhaust stack, h = height of building.

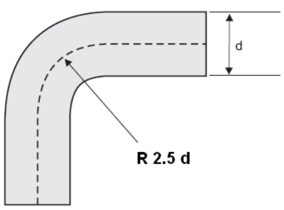
Minimum pipe inside diameter 150 in mm.

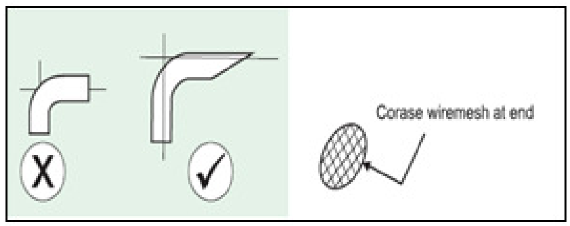
Note :

- If the extension of exhaust pipe is more than 5 meters above Genset, increase pipe diameter by 13 mm for every 3 meters.

Exhaust bend

Bends used in exhaust piping should have a **long smooth bend** as shown in figure to ensure smooth flow of exhaust gases.





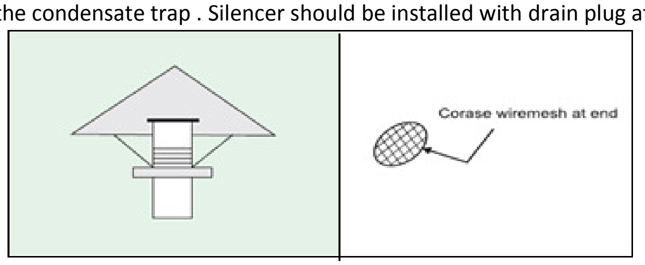
Note :

- **If No. of bends are more than 4** , increase the diameter of exhaust pipe and bend by 13 mm for addition of each bend .
- **The exhaust outlet should be in the direction of prevailing wind**.
- **Exhaust outlet should have 30° cut** at the tail end Coarse wire mesh.

Rain Cap

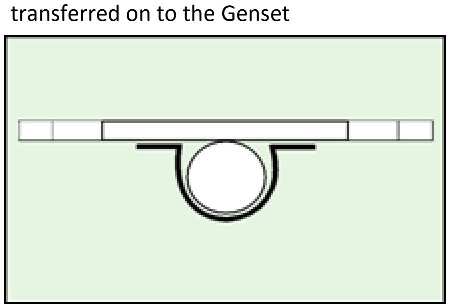
If horizontal exhaust outlet as recommended above is not possible , provide a rain cap over exhaust pipe end as per the figure below . The distance between exhaust pipe and rain cap should be higher than diameter of pipe (minimum 2.5 D) with coarse wire mesh.

- **It is also recommended that horizontal run of exhaust piping should slope downwards** away from engine to the condensate trap . Silencer should be installed with drain plug at bottom.



Exhaust Support

Proper support is required for the extended exhaust pipe above Genset to ensure no self weight is transferred on to the Genset



Type of Exhaust Pipe

Use black MS ERW (Electric Resistance Welded) - Class B Type pipes. Galvanized Iron pipes should not be used. Exhaust pipes should be joined through flanges.

Exhaust pipe through out should be cladded by mineral wool with aluminum sheet (0.8mm). Minimum thickness of cladding should be 50 mm.

Liquid Cooled Engine

KOEL Liquid Cooled Engines use premixed coolant for removal of heat generated in the engine Due to internal combustion & to maintain optimum operating temperature for efficient engine operation**.**

Lub Oil System

KOEL engines utilize the pressure lubrication system with the lub oil filter adequately designed to handle the flow of lubricating oil at the designed pressure and temperature for Lubrication, Cooling and Cleaning of internal components. KGPL Gensets are delivered with initial fill of Lub Oil "K-Oil Super".

- **Check the Lub Oil level with the help of Dipstick to "H" Mark** and top up if required with **"K-Oil Super"** only.

K-Oil Super **SAE 15W 40** is a specially blended engine oil which ensures :

- Increased change intervals. • Better oxidation stability and hence less top-ups. • Reduced frictional wear.
- Increased internal cooling. • Protection against rust and corrosion. • Reduced carbon deposits.

Lub Oil System

⚠ **Use of the wrong grade of oil could lead to**:

- Overheating of the engine. • Sluggish performance. • Excessive fuel consumption
- Increased wear of bearings and other moving components.

Fuel System

The fuel (**High Speed Diesel**) should comply with the International Standards.

DIN 51601 , IS : 1460 1995 , ASTM D 975-81: 1D & 2D

- Ensure fuel filling point is accessible and clean all the time. • Ensure all fuel piping is leak free.
- Avoid fuel spillage • **E**nsure fuel dilution does not occur.• Care should be taken while filling of tank to prevent ingress of dust &moisture inside diesel tank.
- It is recommended to fill fuel after stopping of the Genset to avoid condensation .

⚠ It is recommended not to fill fuel during running of Genset

Electrical System

Earthing

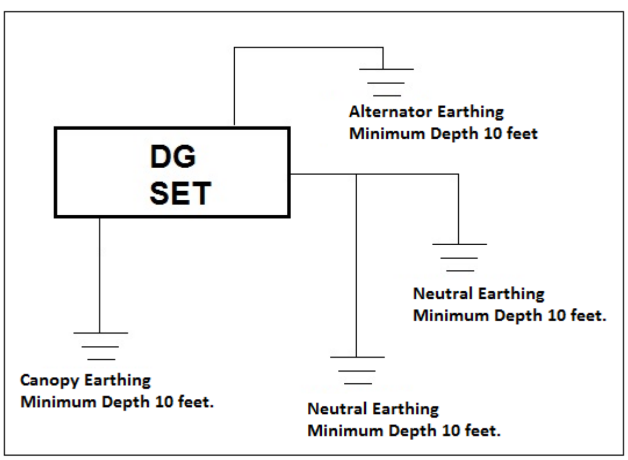
Any leakages in current will be earthed through the shortest route in the link.

KGPI Genset should be connected to earth in accordance with local regulations.

The size of the link used for the main earth connection should be of adequate size as shown in table below.

The generating set and all associated equipment must be earthed before the set is put into operation. 4 numbers of **Earth pits are required as per Indian Electricity rules / local electricity regulations**.

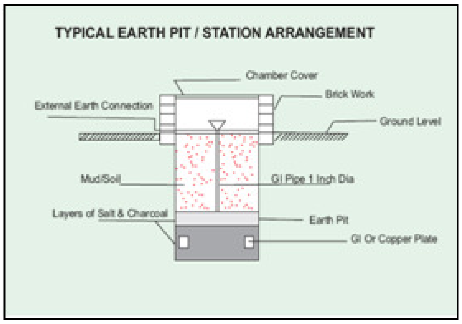
- ➢ 2 Earthing pits for Genset / control panel body
- ➢ 2 Earthing pits for neutral.



Resistance between 2 earth pits should not be more than 5 Ω

Typical Earth Pit diagram is as shown in figure.

Genset Rating	Recommended Earth Strip
625 kVA	50 x 6mm Copper



Power Cable

Cabling from Alternator to Panel should be done with proper size of cable and lugs.

Typical cable sizes provided in the table are indicative . Please refer to cable manufactures for details.

- **Ensure tight crimping / terminations** to avoid overheating and burning of the cable and terminals.
- **Ensure correct size of gland & gland plates** are used to avoid tension on alternator terminal , terminal box and control panel terminations.
- **Ensure phase colour coding** is adopted while terminating cables at both the ends of alternator and control panel.
- **Ensure proper bending radius** is given to the cable to avoid excessive tension on cable termination. For extended control cabling use 10/12 core 2.5 sq.mm. of armoured copper cable.
- **Any type of control & monitoring cables should be sheathed** and should be within 5 meters length of accurate performance.
- **Ensure that the weight of cables does not rest on alternator terminations** . Proper support to be provided from the ground separately through cable tray.

Typical Cable Sizes for KGPL Genset

Cable selection chart:

3-Phase Genset			
Genset Rating	Amp Rating @0.8 pf	Cable Sizes in Sq mm	
		Copper Multistrand	Aluminium
625kVA	869.52	400	300 x 3

Cabling- Polycab Make :

500 kVA	3.5 core x 300 sq.mm Aluminium armoured

Important Notes:

1. **Please use Comet makes cable glands**.
2. Type of cable gland: Single Compression Heavy Duty Cable Gland.
3. Material of cable gland: Brass.
4. The amp. rating of cable is considered as per duct laying method and 700 C operating Temperature mentioned in manufacturers catalogue.
5. Considered amb temp: 40° C.
6. **For grouping of cables** (multiple runs) the derating factors are considered as per duct laying method of cable.
7 **The distance between two parallel cables is considered 30 mm for duration**.

Changeover Specification:

Changeover Specifications		
625 kVA	1000 AMPS	QTY - 1 No.

Battery

Engine model	Ambient Temp	Battery Specification		Battery cable size.	
		12V Starting	24V Starting		
DV12TA G2	- 20 to +10	-	2 x 12V 200 Ah	70 (sq mm)	Multi strand PVC flexible Red and Black colour 1.25 mts copper cable with suitable lugs (one side battery lugs and one side round lugs)
	>+10	-	2 x 12V 180 Ah		

Ensure that the polarity of the battery connections are correct before switching on the DC controls.
Never attempt to start the Genset if battery charge is insufficient to start the engine.
Get the battery charged or replaced and then try to start.
During installation and maintenance, disconnect the battery.
For prolonged storage, used static battery charger to keep battery fully charged.

Specific Gravity of Lead Acid in the Battery at various stages:

- 100% Charge – 1.27 to 1.24 gm/cc (Yellow)
- 70% Charged – 1.24 to 1.20 gm/cc (Blue)
- Needs Recharge – Below 1.20 gm/cc (Red)

Starting and Stopping A New KGPL Genset

Lubrication:

- **Check Lub oil level** with help of Dipstick and if required, top up to H-Mark with “K-Oil Super”.

Fuel:

- **Ensure fuel tank is clean** from inside.
- **Fill fuel tank with clean High Speed Diesel**.
- **Bleed the fuel system** as shown in maintenance manual.

Cooling:

Liquid Cooled Engines:

- Check Coolant level inside Radiator and compensatory Tank . If required Top up with “K –Cool Super Plus “ .

Electrical:

- **Check Battery Polarity.**
- **Check Battery cables are correctly connected and secured.**
- Check all other electrical connections and tightness.
- **Ensure Emergency Push Button is in released condition.**
- Ensure Load Change Over Switch / Breaker is in Off condition.

Procedure: Starting KGPL Genset

Before Starting

- Clean the Genset with dry cloth.
- **Check the lub oil, coolant and fuel levels** – top up if required as per recommendation given.
- **Check for V-belt, Hoses and external leakage if any.**
- Ensure Load Change Over Switch / Breaker is in Off condition.

On Starting

For Key Starting

- **Do not crank engine for more than 5 seconds.**
- As soon as engine fires, release the start key. Leave ignition switch in **“ON”** position.
- If engine fails to start, wait for a minute and try again.
- **Check for leakages if any**.
- **Allow the engine to idle for 3 minutes** and then start loading the Genset gradually in steps.
- **Check for Engine parameters** like Oil Pressure and Battery charging rate at regular intervals while the Genset is in operation and maintain log book.
- **Check for exhaust smoke**.

- **Check for abnormal noise** if any.
- **Check for A C Alternator parameters** like Voltage, Ampere, KW, PF on all phases.
- **Ensure load is balanced** in all phases as per recommendation and within rated output of Genset.
- **Check all gauges** and meters provided in Control Panel are working.

Procedure: Stopping KGPL Genset

Before stopping

- Remove the load from Genset and let it run for about 3 minutes.
- For Key Start, put Ignition key in “Off” condition.

After Stopping

- Check for leakages if any.
- Top up the Fuel Tank.
- Ensure proper closing of Genset Enclosure

LOG SHEEET (Sample for use)

The user is requested to maintain a separate log book registering the following mentioned parameters in the table. Following tables are meant for providing a reference to you .

Date						
Daily use in hrs.						
Hour meter reading						
Cooling Water Temp (^{0}C)						
Lub. Oil	Pressure (Kg/cm^2)					
	Temp (^{0}C)					
Fuel added (ltr.)						
Lub. Oil added						
Engine (rpm)						
Current (Amp) 3-phase						
Voltage						
Frequency (Hz)						
PF & kW if provided						
kWH meter reading if provided						
Remarks – Record events of maintenance/repairs						

Maintenance Tips

First oil change @ 50 hrs or 7 days of commissioning.

Series	Oil Change	Oil Filter	Fuel Filter	Air Cleaner element	V Belt Change
	hours	Change hours	Change hours*	change hours**	hours
DV	500	500	500	500- for dry type air cleaner element. 250- Oil change for Oil bath Air Cleaner	2000

* Fuel Filter Pre and Micro to be replaced alternatively after 250 hrs. for 1.1 Ltr.

** Air filter change is based on Ideal site condition, however extreme dusty condition change frequency is based on occurrence of red band in restriction indicator.

It is recommended to get the routine maintenance done through KOEL Authorized dealer.

Cleaning and Preservation of KGPL Genset

- **When the Genset is kept idle for more than 6 months**, it is advisable to protect it from atmospheric moisture, condensates, internal corrosions and rodents.

Recommended Preservatives:

Manufacturer	Engine lub oil and Fuel system	Engine cooling system for liquid cooled	Unpainted ferrous metal parts
Bharat petroleum	Bharat Preserve oil 30	Bharat Sherol B Emulsion with water ratio 1:20	Bharat Rustrol 152
Hindustan petroleum	Autoprun T120	Koolkit 40.5%Emulsion with water	Rustop 274
Indian oil corporation	Servo Preserve 30 OR Servo 'Run-N' Oil 30	Servo Cut S 20% Emulsion with water	Servo RP 125
Castrol India	–	–	Russtilo DW 904 OR DW901
Veedol tide water oil co.	Veedol 30/40	Veedol Amulkut 4 Emulsion with water in ratio 1:15	Veedol Rustop IT

KGPL - Customer Support Network

- The after sales service to KGPI Genset is provided through KOEL's country-wide network of authorized spare and service dealers. Service dealers are equipped with all necessary tools and tackles to take care of every need of service.
- We understand how important after-sales support is to your business. Where your business operations are far reaching across the country & moreover when they are located in absolute remote places, your business counts on reliability & availability of prompt service to ensure minimum downtime.
- We are committed to provide adequate after sales service support & believe in creating an enduring relationship with every individual customer we serve.
- Our service team is always round the corner to provide onsite assistance and comprehensive Service support ensuring your power needs are fulfilled
- Before commissioning the KGPL Genset, please go through the contents of the warranty document.
- For any of your service requirements please contact nearest service dealer or KALA Help Desk.

Kala Genset Pvt Ltd also provide installation services at site.

– In view of continuous product updating and design changes, all above specifications and dimensions are subject to change without prior notice. Also refer to latest product catalogue for updates.

– For the site conditions other than standard operating conditions, consult KALA or KOEL's Area Office.